M1 Data Mining Project Proposal: Analyzing Reader Patterns and Genre Associations in the Goodreads Dataset*

* CS 4412: Data Mining

1st David Holland
Student at Kennesaw State University
Marietta GA
dholla36@students.kennesaw.edu

***Abstract*—This proposal outlines a data mining project focused on the 2017 Goodreads dataset. This study aims to discover relationships between book genres, user behaviors, and how readers structure their reviews. We will be applying different techniques such as Association Rule Mining, K-Means Clustering, and Latent Dirichlet Allocation. This proposal seeks to understand how readers associate different genres in an attempt to define different segments of the reading community.**

***Index Terms*—Data Mining, Goodreads, Association Rules, Clustering, Text Mining, Pattern Discovery**

I. Introduction

The digital age has transformed reading from a personal, solo activity into a social activity. Various platforms like Goodreads and Fable provide a massive database of user-generated data through ratings, reviews, and custom shelves. While many similar projects exist to predict user's rating there is more value in understanding how users engage with literature. This project focuses on the UCSD Goodreads dataset to find these patterns.

II. Description

This project will utilize the Goodreads Book Datasets (2017) curated by Julian McAuley at UCSD, Rishabh Misra and Ndapap Nakashole

A. A. Source and Scale

- **Source:** https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html
- **Size:** The full dataset contains metadata for 2.36 million books and 15.7 million detailed review texts. For this project, a subset (e.g., the "Young Adult" or "History" category) will be utilized to maintain computational feasibility while ensuring a sample size of over 100,000 records.
- **Format:** Data is provided in the CSV and json format, requiring significant preprocessing and flattening.

Identify applicable funding agency here. If none, delete this.

B. Key Features and Schema

The dataset is split into two primary components:

1) **Metadata (*goodreads_books.json*):** Contains `book_id`, `title`, `authors`, `average_rating`, and importantly, `popular_shelves` (user-defined tags).
2) **Reviews (*goodreads_reviews.json*):** Contains `user_id`, `book_id`, `rating`, `review_text`, and `n_votes`.
3) **User Interactions (*goodreads_interactions.csv*):** Contains `user_id`, `book_id`, `is_read`, `rating`, and `is_reviewed`.

III. C. Data Quality and Preprocessing

Known issues include many user-defined shelves that share common tags like "to-read" and varying review lengths. This is most likely caused from Goodread's ability to easily add books to user's "to-read" shelve. Preprocessing will involve tokenization, stop-word removal for text mining and transforming the nested JSON shelves into a transaction-style for association mining.

IV. Discovery Questions

This project moves beyond prediction models to investigate the following questions the team had:

- What are the hidden associations between specfic book genres and user-defined "shelves"? (Do readers of "Romantasy (Romance Fantasy) also show high interset in "Historical Romance" or other fantsay genres?)
- Can we identify distinct Reader Personas based on interaction metrics? By clustering users based on their average rating variance, review frequency, and review length our goal is to discover if it is possible to classify readers as a critical reviewer or casual consumer.
- What type of patterns emerge from highly-voted reviews compared to low-voted ones?

V. Planned Techniques

The analysis will follow a multistage data mining pipeline as visualized in Figure 1.

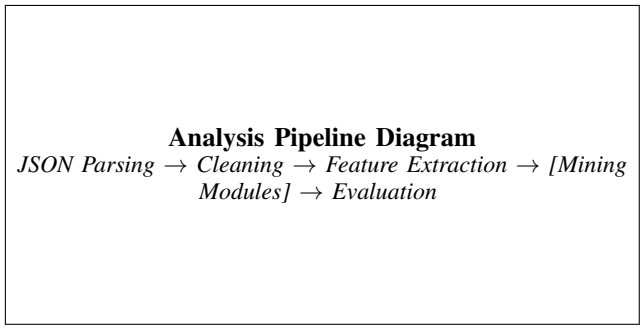

Fig. 1. Planned Data Mining Workflow.

A. Association Rule Mining (FP-Growth)

We will treat each user's shelves as a transaction. Using the FP-Growth Algorithm to discover rules with high lift and confidence to identify non-obvious genre relationships. Our goal is to move beyond the standard "Fantasy" label and discover more sub-genre patterns.

B. Clustering (K-Means)

To answer the second discovery question, we will apply K-Means clustering on standarized numerical features derived the user dataset. We will be using the "Elbow Method" and "Silhoutette Score" to determine the optimal number of reader segments.

C. Text Mining (LDA - Topic Modeling)

Using the Latent Dirichlet Allocation (LDA) algorithm, we will perform topic modeling on the `review_text`. This will allow us to discover the primary topics of discussion within different genres without pre-defining what those topics are.

VI. Preliminary Timeline

The project will proceed according to the following milestones

TABLE I
Project Milestone Timeline

Milestone	Period	Key Activities
M2	Weeks 8	Applying KDD process and Association Rule Mining
M3	Weeks 11	Clustering Analyses, Classification Techniques, detecting Anomalies, and Communicating results
M4	Weeks 14	Text Mining (LDA), Final Report

A. Anticipated Challenges

- Our primary challenge is lack of experience in analyzing data sets.
- Another challenge is the volume of data. The Goodreads dataset is comprised of multiple csv and jsons files that is multiple gigabytes compressed. Handling this in local memory will require using pandas chunking or focusing on smaller genre subsets. Text data can also be "noisy" and will require filtering to produce meaningful topics.
- Due to Goodreads' global reach our team expects to encounter non-English reviews, books, and shelf titles. Our team will have to strip any non-English text out of the datasets in order to prevent any of our proposed tests from failing.

VII. Milestone 2

This section details the work done for Milestone 2 of this data mining project.

A. Preprocessing and Data Transformation

In order to transition from the introductory and theorteical discovery questions from the previous sections our team implemented multiple scripts to select and gather the raw data from the dataset. To address the first discovery question regarding the hidden associations we wrote some python scripts to filter the massive dataset. These scripts took a while to run our computer. First we performed a keyword search across the "popular_shelves" attirbute of the `goodreads_books.json.gz` file. We spefically search for genres: 'romantasy', 'romantic-fantasy', 'fantasy-romance', 'romance', 'fantasy', and 'romantic'. The filtered results are stored in `01_filteredbooks.json`. The resulting json consists of records: book_id, title, average_rating, shelves_list.

To answer discovery question 2, we need to examine the different user interactions, so we implemented a second filtering script utilizing the book_id from our previous script to extract reviews from the `goodreads_reviews_dedup.json.gz` dataset, storing the output in `03_extractedreviews.csv`.

The datasets required significant cleaning before our team could conduct meaningful mining. First, we removed generic administrative tags like 'to-read' and 'ebooks' from the books dataset to reduce noise. Second, to focus our analysis and ensure data comprehension, we removed any non-English records from the reviews dataset using the Python package `langid`.

B. Exploratory Data Analysis

Before applying our mining algorithms, our team examined the distributions of ratings, community-defined tags (for example users can create virtual bookshelves and give them a name), and review engagement to establish a baseline understanding of the "Romantasy" and other romance genres subset.

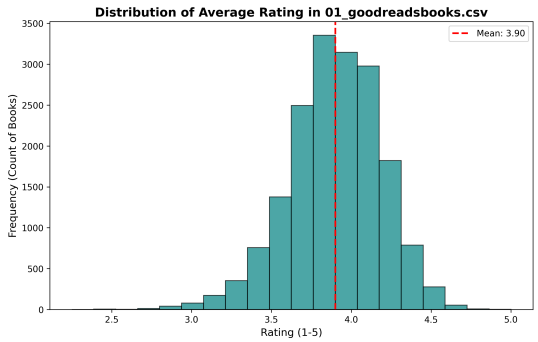

Fig. 2. Distribution of boos rating in 01_goodreadsbooks.csv

Examining the histogram of the average ratings of romance books in the 01_goodreadsbooks.csv we find that most readers rated the books around 4. This suggests that romance readers find books great reads. They often are not critical of books.

1) Shelf Tag Distribution: Analysis of the shelves_list indicates that the tag 'to-read' is the most frequent attribute. During the preprocessing phase, we implemented a filter to only show tags that correspond to actually refer to book descriptions and not generic tags such as "to-read". However after during the postprocessing phase the most prevalent tags included fiction, romance, and fantasy. This aligns with the "Romatansy" genre which confirms Romantasy is the most pollinated genre.

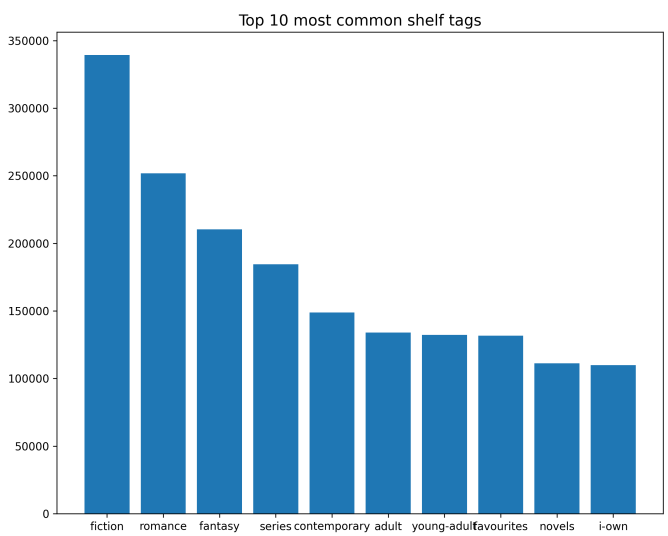

Fig. 3. Distribution of the top 10 most frequent shelf tags post-cleaning

In order to answer discovery question 1 we need to example the different interactions users make so we examine interactions on books across the site along with the types of reviews. We implemented a second filtering script that utilizes the book_id from our previous script to perform an extraction from the `goodreads_reviews_dedup.json.gz` dataset. The extracted data is stored in `03_extractedreviews.csv`.

C. Answering Discovery Question 1: Genre Associations

Our team originally asked if readers of "Romantasy" (Romance Fantasy) also showed high interest in other specific romance sub-genres. After running the FP-Growth algorithm on our transaction data, the top association rules generated lift values of approximately 6.8 to 7.0 for various fantasy sub-genres, rather than romance genres.

While we did not see strong associations between romance genres at our current minimum support threshold of 0.01, our findings partially support the broader hypothesis of sub-genre cross-pollination. Specifically, we demonstrated that readers of fantasy are highly likely to read and tag other closely related speculative fiction genres. This is evidenced by the strong associations found between "paranormal" and "supernatural" fiction, as well as the heavy overlap between fantasy and science fiction readers.

D. Univariate Analysis

Beyond the average ratings, our team examined the distribution of community-defined tags and the volume of review engagement.

1) Engagement Volume:

VIII. Updated Plan for Milestone 3

Moving into Milestone 3, our plan remains aligned with the original Milestone 1 proposal. Since we have successfully established our preprocessing pipeline and validated our association rules, our next focus will be applying K-Means clustering to answer our second discovery question. We will collect user interactions from rating variance and review length—to identify distinct reader categories (e.g., casual consumers vs. critical reviewers). Additionally, we will begin exploring Latent Dirichlet Allocation (LDA) for text mining the review bodies.

Acknowledgment

The author would like to thank Julian McAuley for providing the Goodreads dataset and Kennesaw State University for the computational resources.

References

[1] Mengting Wan, Julian McAuley, "Item Recommendation on Monotonic Behavior Chains", in RecSys'18.

[2] Mengting Wan, Rishabh Misra, Ndapa Nakashole, Julian McAuley, "Fine-Grained Spoiler Detection from Large-Scale Review Corpora", in ACL'19